

Tide retrospective: *2D pure-quartic affine covariance*

Daniel Murfet (with Claude Opus 4.7)

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Abstract

A single Tide-loop excursion in the `laplace` seabed, formalising the affine covariance for the doubly-degenerate 2D potential $L(x, y) = (x^4 + y^4)/24$. The headline theorem `cov_affine_pureQuartic`¹ adds 478 lines of Lean to the seabed; the proof is a near-mechanical mirror of the existing `cov_affine_semiDegenerate`, with the harmonic y -direction replaced by a second copy of the quartic. Zero `sorry`, zero `axiom`, zero `native_decide`; clean Lean diagnostics on first pass after a namespace fix. The interesting friction in this run was not mathematical: two other Tide agents (sextic, bounded-prior) were operating in the same checkout concurrently, and one of them clobbered `Laplace.lean` between `git add` and `git commit`. The fix was a trailing single-line commit; the lesson is to run concurrent Tide agents in separate `git worktree` checkouts.

1 Setting

This Tide step is the fifth excursion outward from the same seabed: the `laplace` repo, which originally formalised the 1D anharmonic asymptotic $\text{Cov}_t[x^2, x] = -2\alpha/(\lambda^3 t^2) + o(t^{-2})$ (Susceptibility Primer, eq. 4.10) and has since been extended by four prior tide steps². This step branches off the resulting `main`, commit 44a6001.

The candidate was picked from a cross-tide candidate survey,³ which catalogued five candidate steps drawn from the wider research portfolio. Of those, the 2D pure-quartic case scored as the cleanest mechanical extension: the existing semi-degenerate 2D file⁴ already provides potential-agnostic 2D wrappers (`partitionFunction`, `gibbsExpectation`, `gibbsCov`) and a worked example of the $\exp(-tL)$ factorisation pattern; the existing 1D quartic file provides closed-form 1D moments. Combining them is a substitution exercise.

The Step 2 deliberation (Claude $\leftrightarrow$ GPT-5.5 Pro) drafted three candidates: A, the affine covariance itself; B, the more general mixed even-power moment family $\langle x^{2m} y^{2n} \rangle_t$; C, just the moment table $\langle z_j \rangle$, $\langle z_1 z_2 \rangle$, $\langle z_j^2 \rangle$. Both parties backed A on the same grounds: A is the smallest statement that closes a useful interface (the affine covariance is what subsequent susceptibility work will call), and B's closed form is mathematically clean but introduces `Real.rpow` exponent

¹In the new file `Laplace/TwoD/PureQuartic.lean`, namespace `Laplace.TwoD.PureQuartic`.

²Branches: `tide/quartic-moments` (1D pure-quartic case), `tide/2d-semi-degenerate` (2D semi-degenerate case), `tide/universal-scaling` (universal scaling identity), and `tide/f1-large-t` (large- t asymptotics of f_1). All four were merged into `main` on 1 May.

³`projects/automation/log/2026-05-02-tide-candidates-survey.md`

⁴`Laplace/TwoD/SemiDegenerate.lean` (existing 2D seabed) and `Laplace/OneD/Quartic.lean` (1D quartic seabed).

bookkeeping (half-integral when $m + n$ is odd) that costs more in Lean than its incremental power justifies. The full deliberation is in the tide log.⁵

2 The thing formalised

For the centred quartic-quartic 2D potential

$$L(x, y) = \frac{x^4}{24} + \frac{y^4}{24}$$

and affine observables $\varphi(x, y) = a_1x + a_2y + c$, $\psi(x, y) = b_1x + b_2y + d$, against the Gibbs measure $e^{-tL(x,y)} dx dy / Z(t)$ on $\mathbb{R}^2$:

For all $t > 0$,

$$\text{Cov}_t[\varphi, \psi] = (a_1b_1 + a_2b_2) \cdot \sqrt{24/t} \cdot \frac{\Gamma(3/4)}{\Gamma(1/4)}.$$

The Lean signature:

```
theorem cov_affine_pureQuartic {t : ℝ} (ht : 0 < t)
  (a a b b c d : ℝ) :
  gibbsCov pureQuarticPotential t
    (fun z : ℝ × ℝ => a * z.1 + a * z.2 + c)
    (fun z : ℝ × ℝ => b * z.1 + b * z.2 + d) =
  (a * b + a * b) * Real.sqrt (24 / t) *
    Real.Gamma ((3 : ℝ) / 4) / Real.Gamma ((1 : ℝ) / 4)
```

in the new file `Laplace/TwoD/PureQuartic.lean`, namespace `Laplace.TwoD.PureQuartic`.

The statement is the doubly-degenerate analogue of the existing `cov_affine_semiDegenerate`: where the latter has one harmonic direction (rate t^{-1}) and one quartic direction (rate $t^{-1/2}$), the present theorem has both directions degenerate and a single rate $t^{-1/2}$. The constants c, d drop out (covariance is shift-invariant in each argument); the cross terms a_1b_2 and a_2b_1 vanish because $\langle z_1z_2 \rangle = 0$ by parity in either factor; what survives is the diagonal contribution $(a_1b_1 + a_2b_2)\langle z_j^2 \rangle$, with $\langle z_j^2 \rangle = \sqrt{24/t} \Gamma(3/4) / \Gamma(1/4)$ from the 1D quartic moment lemma.

Numerical sanity check: `scipy.integrate.dblquad` over $[-6, 6]^2$ agreed with the closed form to 1.4×10^{-14} across four (a_i, b_j, c, d, t) test points (and analogously to machine precision for candidates B and C; all are recorded in the tide log).

3 Proof strategy

The Boltzmann factor $\exp(-tL(x, y))$ separates as

$$\exp(-t(x^4 + y^4)/24) = \exp(-t \cdot x^4/24) \cdot \exp(-t \cdot y^4/24),$$

so the 2D Gibbs measure is the product of two copies of the 1D quartic Gibbs measure. The proof rides this factorisation throughout. The load-bearing Mathlib lemma is `integral_prod_mul`⁶: under no hypotheses, the integral over $\mathbb{R}^2$ of a separable function $f(x)g(y)$ equals the product of the 1D

⁵`projects/primer/tide-log/2026-05-02-tide-2d-pure-quartic.md`

⁶In the `MeasureTheory` namespace.

integrals (and is 0 on both sides if non-integrable, by Bochner convention). Applied to the partition function this gives `partitionFunction_factor`: $Z_{2D}(t) = Z_{\text{quartic}}(t)^2$. Applied to monomials it gives `integral_pow_pow_factor`:

$$\int_{\mathbb{R}^2} z_1^m z_2^n e^{-tL(z)} dz = \left(\int_{\mathbb{R}} x^m e^{-tx^4/24} dx \right) \left(\int_{\mathbb{R}} y^n e^{-ty^4/24} dy \right).$$

The 1D factors are then `quartic_moment_even` and `quartic_moment_odd` (zero by parity).

From there the proof is mechanical: the means $\langle z_j \rangle = 0$ and the mixed first moment $\langle z_1 z_2 \rangle = 0$ vanish via the parity factor; the second moments $\langle z_j^2 \rangle$ reduce via `quartic_expected_value_sq`; the headline covariance follows by expanding $(a_1 z_1 + a_2 z_2 + c)(b_1 z_1 + b_2 z_2 + d)$ and using linearity of `gibbsExpectation` (which itself unfolds against `integral_add` and `integral_const_mul`, having first established 2D atom integrability for each $z_1^m z_2^n e^{-tL}$ via `Integrable.mul_prod` from two `quartic_integrable_pow_pot` witnesses).

The proof body of `cov_affine_pureQuartic` is a six-term `integral_add` cascade (six monomial cross-terms in the expansion of the product of the two affine observables) followed by a `field_simp` and `ring` closer. This is verbatim the same skeleton as `cov_affine_semiDegenerate`; the only changes are the potential, the namespaces, and the absence of the harmonic-direction integrability helper.

4 Roadblocks and resolutions

Namespace clash with the existing 2D excursion. The first build attempt produced a wall of *has already been declared* errors. The cause was straightforward: the existing semi-degenerate file put its helper lemmas⁷ at unqualified names inside `namespace Laplace.TwoD`—under the same generic names that the present file naturally wants for the quartic-quartic potential. With both files in the same namespace, the lemma-name space collides.

The cheap fix (taken here) is to put the present file’s lemmas in a nested sub-namespace `Laplace.TwoD.PureQuartic`; the bare identifiers `partitionFunction`, `gibbsExpectation`, and `gibbsCov` still resolve because Lean searches outward for unqualified names. This is the lighter of the two routes.

The heavier and structurally cleaner route, suggested by GPT-5.5 Pro at the deliberation step and deferred here, is to extract a generic *additively-separable potential* factorisation: prove once that for any $L(x, y) = U(x) + V(y)$ the partition function factorises and expectations of separable observables factorise, then specialise to $U = V = x^4/24$ (this file) and to $U = x^4/24, V = \lambda y^2/2$ (the existing semi-degenerate file). With this abstraction in place, GPT noted, “B becomes almost free”—i.e. the mixed even-power moment table $\langle x^{2m} y^{2n} \rangle$ falls out without further work. The decision to defer was deliberate: the Tide skill prefers small finished steps to medium-sized refactorings, and the namespace pressure is manageable at two 2D potentials. It will not be at three.

Concurrent-agent staging race. The repository hosting this Tide step had two other Tide branches running simultaneously⁸, all operating in the same checkout. Multiple times during this run, the working-tree copy of `Laplace.lean` was overwritten by another agent who was also adding an import for their new file. Once, the overwrite happened in the window between `git add Laplace.lean` and `git commit`—the staged content survived, but Git’s index appears to have

⁷For example, `partitionFunction_factor` and `gibbsExpectation_fst_sq`.

⁸The branches were `tide/sextic-moments` (a 1D sextic excursion) and `tide/grammar-precursor-bounded-prior` (an in-progress excursion toward the grammar paper).

been touched again before the commit fired, with the result that the new file’s `import` line did not make it into the first commit (1e8954b). The working tree then showed `Laplace.lean` as unmodified relative to the new `HEAD`, so a quick `git diff` did not surface the problem; only inspecting the commit’s file list revealed it.

The fix was a single-line trailing commit (efafbfa) wiring the import. This worked but is not a satisfying mitigation. The operational lesson, written up under *Lessons* below, is that concurrent Tide agents should run in separate `git worktree` checkouts so that their working-tree edits cannot interleave.

Beyond these two, no proof-side roadblocks: the proof typechecked clean on the first compile after the namespace fix, with zero warnings. The candidate had been picked precisely so that the seabed’s machinery would already cover the moves; the run paid the cost of that prediction.

5 What was Mathlib, what was new

Mathlib provided. `integral_prod_mul` for the partition-function and moment factorisations. `Integrable.mul_prod` for promoting two 1D integrability witnesses to a 2D one. `Real.exp_add` for the Boltzmann factor decomposition. The standard Bochner integral lemmas `integral_add`, `integral_const_mul`, and `integral_neg` for the affine-observable expansion. `mul_div_mul_left` and `mul_div_mul_right` for the Z_{quartic} -cancellation in the second-moment lemmas.

The seabed provided. This step did not consume new Mathlib API; everything not shipped by Mathlib was already in the seabed. From the 1D quartic file, the partition function and its positivity, the even and odd full-line moment integrals, the second-moment expected value, and a polynomial integrability witness against the quartic Gibbs weight.⁹ From the existing 2D semi-degenerate file, the potential-agnostic `partitionFunction`, `gibbsExpectation`, and `gibbsCov` wrappers (which this file reuses unchanged) and the proof skeleton that this file mirrors.

What was new in this tide. The 2D pure-quartic potential `pureQuarticPotential`, its Boltzmann factor decomposition, five moment lemmas (the two means, the mixed first moment, and the two second moments¹⁰), a private 2D atom integrability witness, and the headline `cov_affine_pureQuartic`. There is no novel mathematical content; what is new is the substitution-into-the-template that gives a working Lean proof.

6 Lessons

- *Concurrent Tide agents need filesystem isolation.* The `Laplace.lean` import-list is a contention point because every new top-level file edits it. Two agents in the same checkout will race; it is not a question of careful sequencing. The structural fix is to run each Tide agent in its own `git worktree` attached to its own branch. Worth lifting to the Tide skill as a preflight step alongside the existing one-tide-branch-per-repo rule.
- *Generic lemma names in the master 2D namespace are a smell.* The existing partition-function factorisation lemma reads, by its short name, as if it were a property of the namespace—but it is in fact specific to one potential. A second potential can only coexist by

⁹Specifically: `quartic_partition`, `quartic_partition_pos`, `quartic_moment_even`, `quartic_moment_odd`, `quartic_expected_value_sq`, and `quartic_integrable.pow_pot`.

¹⁰Named `gibbsExpectation.X` for $X \in \{\text{fst}, \text{snd}, \text{fst_mul_snd}, \text{fst_sq}, \text{snd_sq}\}$. The first-moment lemmas are visibly symmetric, but Lean wants both because of the asymmetric structure of the underlying product space.

sub-namespacing or by suffixing every lemma; neither scales. The right time to extract a generic *additively-separable factorisation* is now (two potentials), not later (three or four). This is the deferred-but-already-cost-positive refactor noted in the follow-ups.

- *The substitution template is robust at one more exponent.* This step confirmed that the $\exp(-tL_1) \cdot \exp(-tL_2)$ factorisation ports cleanly across the quartic-quartic and harmonic-quartic boundary. The next test is whether it ports across the sextic side, courtesy of the concurrent sextic Tide step¹¹; if so, the abstraction is genuinely the right shape.

7 Follow-ups

- *Generic separable-potential factorisation.* For any $L(x, y) = U(x) + V(y)$, a single file proving the partition-function factorisation, the separable-observable factorisation $\langle f(x)g(y) \rangle = \langle f \rangle_U \langle g \rangle_V$, and optionally the bilinear-symmetric-affine-shift algebra of `gibbsCov`. Specialises to both the semi-degenerate and pure-quartic cases; candidate B (mixed even-power 2D moments) drops out for free. Best opened as a cleanup PR after the next 2D Tide step lands.
- *Candidate B harvesting.* The mixed even-power moment table $\langle x^{2m}y^{2n} \rangle_t = (24/t)^{(m+n)/2} \cdot \Gamma((2m+1)/4)\Gamma((2n+1)/4)/\Gamma(1/4)^2$ is now reachable in ~ 30 lines from the present file (or in ~ 3 lines from the separable-potential abstraction above). A small future Tide could harvest it; it is more naturally a corollary of the abstraction than a standalone target.
- *Quartic-sextic mixed potential.* $L(x, y) = x^4/24 + y^6/720$, with rates $t^{-1/2}$ and $t^{-1/3}$ respectively. Mechanical given the present file plus the concurrent sextic seabed. Worth a Tide step once that file merges.
- *Worktree-isolated Tide runs.* Process change rather than maths: prepare a small wrapper that creates a `git worktree` for each Tide branch and runs the Tide agent against that worktree. Avoids the staging race observed in this run.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the `Real.rpow-vs-Nat.pow` caveat for candidate B and the deferred separable-potential refactor suggestion. The full consult and the tide log itself (with candidate drafts, vote, numerical check, and result) are preserved in the primer tide-log.¹²

¹¹The 1D sextic file `Laplace/OneD/Sextic.lean` from `tide/sextic-moments`.

¹²GPT response: `projects/primer/tide-log/gpt55.tide_2d.pure-quartic-v1.md`. Tide log: `projects/primer/tide-log/2026-05-02-tide-2d-pure-quartic.md`.